waveguard touch User Manual

Dry EEG cap for eego amplifiers

Revision 4

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Important disclaimer

Every care has been taken to ensure the accuracy of the information in this document. However, we assume no responsibility for errors, inaccuracies, or omissions that may appear in this document.

We reserve the right to change the product without further notice to improve reliability, function or design. This document is provided without warranty of any kind, implied or expressed, including fitness for a particular purpose.

Please read this manual carefully before using a ***wave****guard touch* dry EEG cap.

The cleaning and disinfecting procedures as described aim at a most careful cap maintenance. However, eemagine GmbH cannot control whether these procedures are properly followed nor can we assess the quality of third-party cleaning products. As a consequence, eemagine GmbH does not assume any liability for the cleanliness of caps and the fitness of caps for EEG recordings after cleaning. In particular, eemagine GmbH does not assume any liability for infections that may be caused by using EEG caps that are not previously washed and/or disinfected by the user.

In addition to this information, please always follow the regulations that apply locally in your country and hospital or institute. Contact your distributor if you are in doubt about what regulations apply and how to meet them.

Please contact your distributor in case of a defective cap or other required support. Do not try to repair caps yourself. If you do so, any warranty will be void. Any serious incident that has occurred in relation to the device should be reported to the manufacturer and the competent authority of your country.

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1. Safety information

1.1. Regulatory information for Users in the USA

waveguard touch EEG caps (product code CY-xxx) are intended for research purposes only and are not included in the 510(k) clearance for other cap variants (product codes CA-xxx / CS-xxx).

1.2. Regulatory information for Users in Canada

waveguard touch EEG caps (product code CY-xxx) are intended for research purposes only and are not included in the Medical Device License (MDL) for other cap variants (product codes CA-xxx / CS-xxx).

1.3. Intended Use

waveguard EEG caps are intended for recording of electroencephalographic (EEG) signals, including event-related potentials (ERP) and evoked potentials (EP) on subjects or patients with intact scalp in combination with medical-grade EEG recording equipment (CE class IIa certified inside the European Union; class II device according to FDA regulations for USA).

The cap must be dry and clean when used. The application of caps on subjects should only be carried out by medically trained personnel (EEG technician). The technician must remain present during the entire recording session to prevent subject risks such as strangulation. Special infection control requirements may apply.

***wave**guard touch* EEG cap is a medical device of CE class I according to the Regulation (EU) 2017/745 (MDR) and is also in conformity with 2011/65/EU (RoHS). ***wave**guard touch* EEG cap is denoted with the MD symbol in the device label to indicate it is a Medical Device.

1.4. Intended subject population

A. Age: 2 yrs. old and above. Head-size (circumference): 38 to 60 cm.
B. People without disabilities in expressing themselves and communicating their needs.
C. People without any temporary or constant increased skin sensitivity.
D. People without restrictive hair style (e.g., dreadlocks, voluminous or styling products).
E. Subject is instructed to move as less as possible while wearing the cap.
F. Subject is awake when applying/removing cap.
G. Subject not in laying position.

1.5. Intended user

The operator must be an adult (> 18 years) trained in neurophysiology procedures and must have experience in dealing with EEG applications; the range of operators is limited to:

1. Lab technician, medical technical assistant
2. The operator should fully understand the procedures described in the User Manual
3. English, local language proficient if localized User Manual is available.
4. Professional neurophysiological terminology.
5. Experience in electrode positioning on human heads.
6. Experience in working with subjects (research).

1.6. **Contra-indications**

The caps should not be used on subject with skin sensitivity, skin injuries or problems such as blisters, burns, wounds from operation or other superficial wounds or transmissible diseases, such as – but not limited to – hepatitis, human immunodeficiency virus (HIV) or Creutzfeldt-Jacob disease.

In the event of defibrillation, disconnect the cap and/or electrodes from the EEG recording equipment first.

People with increased pressure sensitivity on the head and/or pressure induced migraine should not use the cap.

1.7. **Side effects**

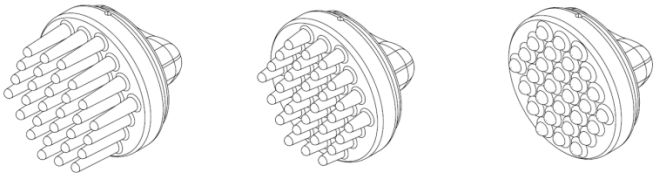
No side effects are known if properly applied. Please apply the correct size of the cap. Keep in mind that a too small cap may cause headache. Slight red marks in the form of dots on the exposed skin (e.g. at the forehead) due to contact pressure are a regular effect of wearing the cap and will disappear within a short time.

2. Important technical notifications

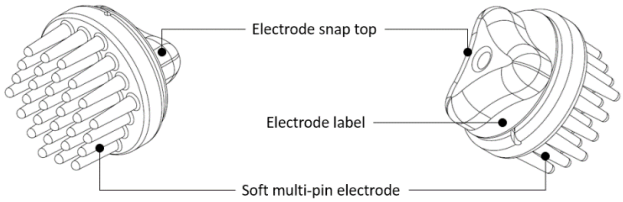
- Do not use incompatible electrode materials like platinum, tin, gold or stainless steel in the same recording.
- Do not use any hard-mechanical tool for cleaning of electrodes. The electrodes may damage.
- Do not put high mechanical stress on the electrode pins, such as squeezing or actively bending.
- Always apply the correct size of cap (related to head circumference).
- Make sure that the cap is completely dry before use.
- Do not keep the cap submerged in water for prolonged periods of time (maximum one hour).
- Do not wet the caps with ionized water or liquids with high salt concentration.
- Do not expose the cap to hot environments over 50° Celsius (or short rinsing at 60° Celsius). The cap fabrics as well as the wiring do not withstand higher temperatures. Hot-air drying may significantly reduce the lifetime of the cap.
- Do not try to wring or squeeze the cap. We recommend to gently dab the cap fabric using a towel, or to hold the cap between two towels for drying.
- Do not use any autoclaving procedure on the cap.
- Avoid any water spilling on the connectors. Keep them in safe distance.
- Please dry the cap ideally in an upside-down position, or on a head model, while the cable being placed higher than the wet cap, in order to prevent adapter and tubing to be contaminated with water.
- Avoid the connectors, dropleads or any other metallic material in contact with the electrodes to prevent corrosion.
- Pull the cap inside-out for drying.
- Always disconnect caps by pulling on the connector housing. Do not pull the cable. Note that dragged-out cables are not covered by the cap warranty.
- Do avoid exposing the electrodes to the sunlight or UV-light during storage.
- The coated Ag/AgCl electrode elements are not suitable for any rechlorification.
- waveguard touch is not suitable for use in MRI or MEG environments.
- Make sure the cap is adequately cleaned prior to disinfecting.
- Do not use any kind of alcohol for disinfection of waveguard touch.
- Do ask the subject about his/her comfort during recordings series.
- Do not apply the cap on a wet scalp. Make sure that the hair of the subject(s) is completely dry before applying the cap.
- waveguard touch may not be suitable for some individuals, especially for those with uneven head-shape or dense hair-growth.
- Make sure that no external pressure is placed on the electrode, for instance by resting the head on a surface.
- Do not attach the cap connectors to the wrong EEG IN inputs of the amplifier.
- Do not push the electrode against the subject without a swipe movement or with high pressure.
- Do not use any electrolyte gel in combination of the waveguard touch dry EEG cap. It may change the electrode property or damage the electrodes.

3. Cap design

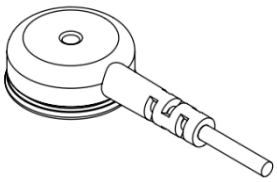
3.1. Cap components



1 - Multi-pin dry electrodes with 30 conductive pins on each electrode



2 - Snap mechanism holds the electrodes in their designated positions on the cap



3 – Flat-snap drop-lead adapter for REF and GND positions

4 - Double-layer protects the cabling; color-coded fabric indicates size

5 - Connectors are compatible with the eego amplifiers (please refer to the datasheet of each cap model for specific compatibility)

3.2. Compatibility

The **wave**guard touch dry electrode EEG caps are designed for the use with ANT eego EEG amplifiers. Different models of **wave**guard touch can either connect to EE-2xx or EE-4xx eego variants. The compatibility of each **wave**guard touch model is described in the dedicated product datasheets.

3.3. Cap features

Feature	*Description*
Soft, Coated Ag/AgCl	**wave**guard touch dry electrodes are passive sensors measuring the small changes in the electrical potential on the scalp. The coated Ag/AgCl electrode has comparable characteristics as the golden standard sintered Ag/AgCl electrodes for the wet caps. At the same time the electrode is soft and adapts well to different pressure forces.
Multi-pin electrode form	Each dry electrode consists of 30 single conic pins with circular tops of 1 mm in diameter. The multi-pin shape enables easy passing through the hair layer and provides sufficient contact surface for each individual electrode.
Graduated electrode length	The height of the electrode pins varies in 3 heights. Long - 6 mm, Medium - 3 mm and Short - 1 mm. Different electrode heights are applied to different electrode positions, in order to prevent peak pressure points on the scalp and ensure optimal fit.
Moisture-reducing fabric	The cap fabric is 83% Polyester, 17% Elastan (spandex), a polyester fiber mesh that supports improved moisture reduction. This fabric also provides good compromise in elasticity and stability. It is soft enough for subject comfort, at the same time applies enough force on the electrodes for the continuous contact with the scalp.
Cable protection	All dry electrodes are mechanically fixated on the cap. The outer and inner fabric layers protect the sensitive cables from damage.
Optimal pressure distribution	The tight cap-fit ensures best contact of the electrodes with the scalp and applies homogenous pressure across all channels.
Up to 64 channels	Current models cover 8 and 64 positions on the scalp. Refer to the datasheet regarding electrode number and layout of each cap model.
Two different layouts	Current models are either in 10/20 or equidistant layout. Refer to the datasheet regarding electrode number and layout of each model.
Active shielding	An actively shielding coaxial layer in each channel wire protects the signal from power line disturbance or other external noise. Any capacitive effects between shield layer and signal wire are optimally reduced by having the shield layer at the same potential as the lead wire.
Optimal performance with eego EEG amplifiers and eego software	In combination with an eego EEG amplifier the potential of the active shields are optimally set. Shielded caps should never be used with amplifiers that do not provide the active shielding. The eego software can support the analysis with an optional method for automatic artifact correction and bad channel detection.

*For further information about specific **wave**guard touch models, please refer to the related product datasheet.*

4. Cap application

4.1. Subject preparation

1. Wash the hair in the morning

Make sure the subject has properly washed the hair, best in the morning before the experiment. This removes the major layer of oil and dead skin residues.

Make sure that the subject did not use any hair conditioner. Conditioners may establish a protective layer on the scalp which could increase the electrical impedance for the electrodes.

✓ Wash the hair on the same day before measurement

✓ Brush the hair before measurement

✗ Do not use hair-conditioner

2. Brush the hair and clean the scalp

It is advantageous to comb the hair using a relatively hard hairbrush before applying the cap. Please be sure to use a comb that avoids static charge. You can either assist the subject with the brushing, or have the person do this him/herself. The brushing removes some dead skin particles/dandruff, resulting in lower electrode impedances.

Try to establish a homogenous distribution of hair density on the subject's head. Untie the hair if necessary. Brush the hair and direct them into a symmetric shape.

3. Preparation of REF and GND positions

The REF and GND is applied at the left and right mastoid bones. Use abrasive skin preparation gel to cleanse the skin surface from any fat and residues of the area where the electrodes will be applied later.

4.2. Cap positioning and fixation

1. Measure the head circumference

Measure the head circumference of the subject with a tape band. Head circumference is measured over the most prominent part on the back of the head (occiput) and just above the eyebrows (supra-orbital ridges). Prepare the correct cap size (according to the size table) for application.

Using the correct size of cap (refer to the head circumference table) is important for achieving a good accuracy of electrode positions. Applying a cap which is too small for the head may cause excessive wear of the fabric and induce pain. A cap which is too big for the head may cause loose electrode contacts and bad channels.

✓ Always apply the correct size of cap (related to head circumference).

✓ Make sure that the cap is completely dry before use.

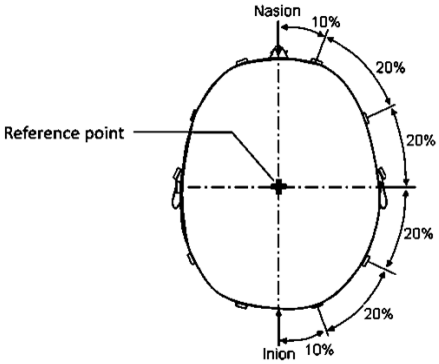
✗ Make sure that no external pressure is placed on the electrode, for instance by resting the head on a surface.

✗ Do not apply the cap on a still wet scalp. Make sure that the hair of the subject(s) are completely dry before applying the cap.

✗ Individual subject may not suit for dry electrode measurement, especially those with uneven head-shape or dense/curly hair-growth.

2. Measure the reference point

Find the reference point halfway the distance of Nasion to Inion and halfway the distance between the two pre-auricular points. Mark the position with an eye-liner or other skin friendly marker.



3. Position the first electrode

For ***10/20 layout***, position the Cz electrode on the marked reference point. Make sure that this electrode position does not shift during the cap application.

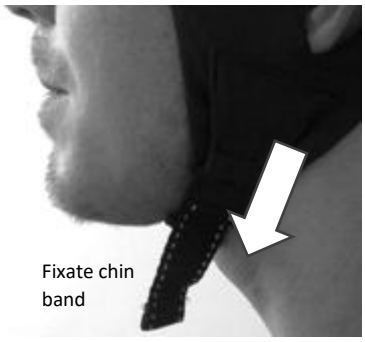
For ***hexagonal equidistant layout***, position the 4Z electrode on the marked reference position. If needed, the equidistant layout can be shifted along the longitudinal line. (e.g. if the frontal or occipital electrodes need to cover lower forehead/back areas). Make sure to note down the distance between 4Z and the marked reference point.

If using the spatial down-sampling function in the eego artefact correction module, it is required to position the 4Z electrode exactly on the reference point measured above.

For detailed layout chart and all positions/naming, please refer to the specific product datasheet of each model.

4. Stretch the cap over the head

Gently pull the cap over the subject's head, starting at the upper central area. Place the cap from the top of the head, starting with the vertex electrodes and then work your way to downward electrodes. Make sure that the cap is applied symmetrically. The fabric should be smooth in all locations after application.



5. Adjust electrodes

Each electrode should have an upright stand on the scalp.

6. Tighten the cap fit

Pull the ear-pieces of the cap downwards to tighten the cap in order to fit on the scalp. Fix the final position of the cap with snapping of the chin band.

7. Apply GND and Reference electrodes

Click the adhesive pad electrodes on the 2 flat-snap drop-leads (GND and REF). Remove the protection layer of the adhesive pad electrode. Position the REF and GND electrodes on the mastoid bones. It is possible to place them under the fabric; however, they should not overlap with close-by dry electrodes. You should remove skin residues with abrasive skin-preparation gel on these positions before applying GND and REF.

✓ Only connect the cap after you finished preparing the subject

× Do not use incompatible electrode materials like platinum or stainless steel in one recording.

4.3. Connecting the cap with the amplifier

The cap comes with a connector for every set of 32 or less electrodes. Make sure that matching connectors are used. Cable outlets at the cap side are numbered if more than one connector is present. When using more than 32 channels, double-check that the numbering is corresponding to the EEG inputs on the amplifier.

When disconnecting a cap, take special care not to pull at the cables but hold the connector housing tightly to remove the plug.

Connect the cap connectors with the correct EEG IN of the amplifier.

✓ Connect each cap connector with the corresponding EEG IN as indicated by the number on the connector label.

× Never drag at the cable. Note that damaged cables due to dragging are not covered by the cap warranty.

4.4. Optimization of electrode contact

1. Check the impedance

Use the impedance measurement tool of the eego software to assess the connectivity of the electrodes. Connectivity is sufficiently established with impedance being below 1 MOhm. However, an impedance below 500 MOhm is recommended. REF and GND should be below 20 kOhm after application.

2. Optimize electrode contact

For channels above this impedance, you can adjust the specific electrode position by gently wiggling the electrode. If a wide range of electrodes are not properly attached, then you can try to wiggle the whole cap with your hand palm. Be careful not to shift the complete positioning of the cap.

Sometimes it is difficult to penetrate the hair layers of the subject. In this case, swipe the electrode from left to right, while pushing on the electrode, in order to push the hair aside.

3. Adjust dynamic range

Please set the dynamic range of the eego amplifier to 750mV when using the waveguard touch cap for your measurement.

✓ Impedance measurement for waveguard touch is possible for eego software 8.0 and higher

✓ Wait at least 5 seconds after applying the cap to read the correct impedance values.

× Do not push the electrode against the subject without a swipe movement or with high pressure.

× Do not use any electrolyte gel in combination of the waveguard touch dry cap. It may change the electrode property or damage the electrodes.

4.5. Electrode performance

1.	Electrode impedance	Connectivity is sufficiently established with impedance being below 1 MOhm. However, an impedance below 500 kOhm is recommended. REF and GND should be below 20 kOhm after application.	
2.	Signal drift and saturation	Signal drift can occur when the subject starts sweating during a recording. In order to prevent saturation, dynamic range of the amplifier needs to be adjusted to 750 mV.	ⓘ Sweat changes the material composition and therefore the electrical properties between electrode and skin. Usually with wet caps, electrolyte gel act as an electrochemical buffer. Without this buffer, influence of sweat is much greater.
3.	Artifact correction	Please refer to your eego software user manual for appropriate application of artifact correction algorithms.	
4.	Global channel reliability	On average 80% of the dry electrodes will provide signal quality equivalent to gelled electrodes.	ⓘ Take care in your application that a percentage of your electrodes might not show signals as expected with a gelled cap.

4.6. Maximum duration of application

Do not apply the cap on the subject for longer than 45 minutes. Depending on individual sensitivity against pressure, subject comfort may be reduced after a certain amount of time.

✓ Ask the subject about his/her comfort during the recording session.

5. Cap Maintenance

5.1. Cleaning

Make sure that the cap is properly cleaned after each measurement. We strongly recommend rinsing the cap with lukewarm water immediately after use.

You can do this at a regular sink - best is to use a showerhead attached to the faucet. Hold the cap inside-out with one hand, and rinse the cap, especially in between of electrode pins with water. You can also clean the cap in a bucket of water. Important note: cleaning is not disinfection, please refer to the next chapter for disinfection procedures.

Using water to clean the cap:

1. Completely submerge the cap in a bucket of lukewarm tap water. Be careful to not wet the connectors.
2. Pull out and re-submerge the cap. Repeat this procedure at least 5 times.
3. Remove the cap from the bucket in order to dry.

Keep the cable and connectors away from the water when cleaning the cap. Avoid any moisture near the connectors in order to completely prevent possible short-circuits or corrosion. In the event that you need to clean the connectors, use a cotton stick slightly damp with alcohol (do *not* use this on the electrodes).

✓ Make sure that the cap connectors always remain dry.

× Do not keep the cap submerged in water for prolonged periods of time (maximum one hour).

× Do not use any hard, mechanical tool for cleaning of the electrodes. The electrode may damage.

× Do not squeeze or wring the cap!

× Avoid any water spilling on the connectors. Keep them in some distance.

5.2. Disinfection

An example of disinfectant solutions that are known to work satisfactorily is Korsolex Extra (by BODE Chemie, Germany). You must follow the disinfectant instructions for use and especially rinse the cap properly afterwards. The list below is an example of the procedure. Several parts, such as labels and the fabric, may bleach slightly. This is normal and does not interfere with the normal usage of the cap. However, the extensive use of disinfectant may reduce lifetime of the caps.

Disinfection solution

all quantities are stated for Korsolex Extra only. For other providers please refer to their guidelines.

1. Mix 4% of the disinfectant liquid with 96% distilled water in a bucket.
2. Wear protection gloves.
3. Completely submerge the cap in the liquid. Be careful to not wet the connectors.
4. Pull out the cap and submerge it again in the solution. Repeat this procedure at least 5 times.
5. Leave the cap submerged in the disinfectant for about 15 minutes.
6. Remove the cap from the bucket and rinse the cap with distilled water.

✓ Make sure the cap is adequately cleaned prior to disinfecting

× Do not use any kind of alcohol for disinfection of **wave**guard touch.

× Do not keep the cap submerged in any disinfection liquid for prolonged periods of time.

Neutral soap

1. Mix 10% of the neutral, non-perfumed soap with 90% distilled water in a bucket.
2. Completely emerge the cap in the liquid. Be careful to not wet the connectors.
3. Pull out the cap and submerge it again in the solution. Repeat this procedure at least 5 times.
4. Leave the cap submerged in the soap solution for about 15 minutes.
5. Remove the cap from the bucket and rinse the cap with distilled water.

✓ Make sure the cap is adequately cleaned prior to disinfecting

× Do not use any kind of alcohol for disinfection of **wave**guard touch.

× Do not keep the cap submerged in any disinfection liquid for prolonged periods of time.

5.3. Drying

The cap has to be completely dry before next use. A solution to shorten drying time is to use a cooling (table) fan. In the cool air stream, drying is quick and safe, within approximately 30 minutes. Place the cap in ca. 50 cm distance to the fan.

Ensure that no remaining water or other liquid may enter the connector during drying.

Avoid the connectors or other metals being in direct contact with the electrodes.

✓ Please dry the cap ideally in an upside-down position, or on a head model, while the cable being placed higher than the wet cap, in order to prevent adapter and tubing to be contaminated with water.

✓ Pull the cap inside-out for drying.

× Do not try to wring or squeeze the cap!

× Do not expose the cap to hot environments over 50° Celsius.

× Do not use a hot hair dryer to dry the cap.

5.4. Storage

Avoid long-time exposure of the cap towards sunlight or UV-light during storage. Store the cap in a dark and dry place (best in a suitcase or plastic bag).

Avoid the connectors or other metals being in direct contact with the electrodes.

5.5. Transportation

Avoid long-time exposure of the cap towards sunlight or UV-light during transportation. The cap should always be transported with the **electrodes facing outside (cap inside-out).** Otherwise the dry electrodes may clinch on each other, thus damage the electrode coating.

✘ Not like this:

✓ Correct:

With the cap inside-out, put all metallic parts in the cap (connectors, dropleads) to avoid direct contact with the electrodes facing outside.

6. Size information

Order code	Sizes	Discription	Head circumference (cm) Min.	Max	Fabric color scheme			Number of electrodes in cap	
								08	64
11	L	Large	55	60	●	●	●	●	●
10	M	Medium	50	55	●	●	●	●	●
09	S	Small	46	50	●	●	●	●	●
08	C	Child	42	46	●	●	●	●	●
07	I	Infant	38	42	●	●	●	●	

(Size indication serves as guideline only – optimal fit depends on head shape and hair style)

7. Product lifetime

How long a cap can be used with all electrodes intact and without compromising signal quality depends upon many factors. Careful handling and thorough cleaning are most important. The electrode sensors, the fabric along with all other parts and materials have been selected and designed such that more than 250 cycles of recording and cleaning can be reached.

8. Warranty and support

The **wave**guard touch EEG cap is made of high-quality materials:

Electrode	Coated Ag/AgCl on Polyurethane (PU)
Cap fabric	83% Polyester, 17% Elastan (spandex)
Electrode housing	Polyurethane (PU)

All caps are carefully inspected before they leave the factory. A **wave**guard touch EEG cap is delivered with a limited warranty for the period of **12 months** after delivery, based on the condition that it was handled according to the indications for use described in this user guide. In summary, the limited warranty ensures that each product will conform to its specification and will be free from defects in material and workmanship, including:

- Loose seams, holes in the cap fabric, e.g. caused by defects in the sewing process
- Broken electrode elements, if not caused by inadequate cleaning/handling
- Broken wires inside the cap, defective cable extending from cap to connector
- Defects inside the cap connector

For repair claims please contact your distributor to get detailed information. Warranty is excluded for damage caused by misuse of the cap and the following examples:

- Broken/missing connector latches (high-density connectors only)
- Defects in the chin-strap/chin-band
- Broken/missing electrode housings
- Normal wear-and-tear of the electrode labels or other labels attached to the cap

Note: final judgement is taken by *eemagine GmbH* after having received the cap in question.

IMPORTANT: Caps returned to the manufacturer for repair – warranty or not – are sent back on the customer's account without inspection/repair if they are not thoroughly cleaned.

9. Waste disposal

Caps must be disposed of as electronic waste in accordance with local legal regulations.

WARNING: Before you dispose of any item, you must ensure that all materials that have been in contact with the subject are not contaminated. You may have to thoroughly clean and sterilize before disposal.

Appendix: EU Declaration of Conformity

Hereby we declare in exclusive responsibility that below identified products fully comply with the regulations of the Medical Device Regulation (EU) 2017/745 (MDR) on medical devices.
To the best of our knowledge given the information provided by our suppliers, products are also in conformity with the Council Directive 2011/65/EU (RoHS) on the restriction of the use of certain hazardous substances in electrical and electronic equipment, as amended by Directive (EU) 2015/863.
Products are not made with natural rubber latex, and do not contain BPA.

waveguard™ EEG cap and accessories

Electrode caps, adapters and leads for recording of EEG signals

The product family consists of devices that are classified according to MDR Annex VIII, rule 1, as **CE class I**, and meets the applicable *General Safety and Performance Requirements* of the MDR Annex I.

We apply the conformity assessment procedure in accordance with MDR Annex IX and issue this declaration of conformity as manufacturer.

Products are marked with CE

The applicable sections of the following standards have been used as a basis for the conformity:

- <IEC 60601-1:2005/A1:2012>
- <EN ISO 14971:2019>
- <EN ISO 10993-1:2009, EN ISO 10993-5:2009, EN ISO 10993-18:2009>
- <IEC 62366-1:2015>
- <EN ISO 15223:2021>

This declaration DOC-QA-0001 is valid for product codes REF as stated in the appended product list.

The corresponding *Global Medical Device Nomenclature* (GMDN) code is: 63131.

We are registered with HIBCC, with *Labeler Identification Code* (LIC): B195.

EUDAMED assigned our *Single Registration Number* (SRN): DE-MF-000005560.

The *notified body* involved in the assessment of our quality management system is:
<notified body name, address>.
Current certificate EN ISO 13485:2016 <registration information>.

Dated at Berlin, Germany, on the _____ day of _____ <2022>.

_____

For the exact <information> in the above text, see the official, signed declaration DOC-QA-0001. The contents of similar conformity statements may be different in countries outside the European Union.